Source Apportionment Techniques

Conventional and Real-Time Approaches

Student reading material accompanying the presentation

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How to use this material: This text follows the slide sequence but is written as a short chapter rather than as slide notes. It can be read before or after the lecture to consolidate the main ideas and terminology.

Learning objectives

- Define source apportionment and explain why it is useful for particulate matter air-quality management.
- Describe the mass-balance idea behind receptor models.
- Distinguish between PCA, CMB, and PMF in terms of purpose, required information, strengths, and limitations.
- Explain how online instruments and automated processing can support real-time or near-real-time source apportionment.
- Recognize why data quality, uncertainty estimation, and model diagnostics are critical for interpreting results.

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1. What is source apportionment?

Source apportionment is the process of identifying the main sources that contribute to measured particulate matter and estimating how much each source contributes. In practice, this means turning an ambient air pollution dataset into source-related information that can be used for scientific interpretation, air-quality management, and policy support.

For PM, the key idea is that different emission sources tend to produce particles with different chemical compositions. Traffic emissions may be associated with elemental carbon, copper, zinc, or other traffic-related tracers. Biomass burning may show potassium, black carbon, and organic carbon. Dust and soil particles often contain aluminium, silicon, calcium, iron, and titanium. Source apportionment uses these differences as chemical fingerprints.

The method is useful because reducing PM concentrations requires knowing where the PM is coming from. A monitoring station may show that PM2.5 is high, but this alone does not indicate whether the main cause is traffic, residential wood burning, dust, secondary aerosol formation, industry, or regional transport. SA helps convert concentration measurements into source-specific evidence.

2. How does source apportionment work?

Most source apportionment methods are based on a mass-balance concept. The measured composition of PM at the receptor site is treated as a mixture of contributions from several sources. The measured data matrix, often called X, contains samples in rows and chemical species in columns. The model attempts to explain X using source contributions, source profiles, and residual error.

A source profile describes the relative chemical composition of a source. For example, a dust profile would be rich in crustal elements, while a combustion profile would include carbonaceous components. A source contribution describes how much mass a source contributes at a given time. Together, profiles and contributions allow the measured data to be reconstructed as a combination of sources.

Different models solve the mass-balance problem differently. Some models require known source profiles before the calculation begins, while others infer profiles directly from the ambient data. In all cases, interpretation requires both statistics and atmospheric knowledge.

3. Conventional versus real-time SA

Conventional source apportionment is usually based on filter sampling followed by laboratory analysis. Filters may be collected daily or over another defined sampling interval, and then analyzed for mass, elements, ions, EC/OC, organic markers, or other species. The advantage is that the dataset can be carefully prepared and evaluated before modeling. The disadvantage is delay: results may become available days, weeks, or even months after sampling.

Real-time or near-real-time source apportionment uses online instruments that continuously measure PM components. Because the data are produced at high time resolution, the source apportionment can be updated soon after measurement. This makes it possible to study diurnal cycles and short pollution events that are hidden in daily averages.

The two approaches should not be viewed as competitors. Conventional analysis is essential for robust interpretation, model optimization, and validation. Real-time analysis is valuable when rapid, source-resolved information is needed for event interpretation, monitoring networks, or operational air-quality management.

4. The model landscape

Source apportionment includes a family of methods rather than a single technique. Receptor models focus on the pollution measured at the receptor site and work backward to infer sources. Source-oriented models, such as chemical transport models, start from emissions and simulate transport, dilution, chemical transformation, and deposition. Chemical reconstruction is another useful approach, where PM mass is reconstructed from measured chemical components.

This presentation focuses mainly on receptor models, especially PCA, CMB, and PMF. These models differ in the amount of prior information they need. PCA is mainly exploratory, CMB requires source profiles, and PMF estimates both profiles and contributions from the ambient dataset. Hybrid methods, such as ME-2, can include partial prior information through constraints.

There is no universally best model. The appropriate choice depends on the question, the available data, the complexity of the site, the chemical species measured, and the expected sources.

5. Conventional receptor models

Receptor models begin with measured ambient composition at a monitoring site. They are called receptor models because they focus on what is observed at the receptor, rather than starting from emission inventories. The aim is to infer source contributions from the chemical fingerprints observed in the measured PM.

PCA, CMB, and PMF represent three different ways of approaching this problem. PCA is a pattern finder: it identifies groups of variables that vary together. CMB is a known-profile method: it estimates source contributions using predefined source fingerprints. PMF is a factorization method: it separates the ambient data into factors that are interpreted as sources or atmospheric processes.

The model output should never be accepted automatically. Source naming and interpretation must be supported by chemical tracers, expected temporal patterns, local knowledge, and diagnostic checks.

6. PCA: what it says and how it works

Principal Component Analysis is a statistical method that transforms many correlated variables into a smaller number of uncorrelated components. In an aerosol dataset, this can reveal groups of species that tend to vary together. For example, calcium, aluminium, silicon, and iron may load strongly on the same component, suggesting mineral dust. Copper, zinc, and elemental carbon may point toward traffic-related sources.

PCA is useful as a first look at the data. It can help identify patterns, outliers, correlations, and possible source groupings before more formal receptor modeling. However, PCA is not designed specifically as a physically constrained source apportionment model. Components may contain negative values, uncertainty is not normally included as a weight, and the components are mathematically constrained to be orthogonal, which may not reflect real atmospheric sources.

Therefore, PCA should be interpreted cautiously. It can suggest source patterns, but it does not by itself provide a complete, physically constrained source contribution estimate.

7. PCA: pattern finder

A helpful way to think about PCA is that it rotates the data into new axes, called principal components, that explain the maximum possible variance. The first component explains the largest share of variability, the second explains the largest remaining share, and so on. These components may be useful for recognizing patterns in the data.

In teaching terms, PCA can be introduced as a map-making tool. It helps the analyst see which species move together and which samples behave differently from the rest. This is particularly useful before running PMF or CMB because it can reveal whether the dataset contains enough variability to separate sources.

The main warning is that PCA components are not automatically equal to sources. A component can represent a mixture of sources, an atmospheric process, or a statistical pattern that has no direct physical meaning.

8. CMB: what it says and how it works

Chemical Mass Balance estimates source contributions by combining measured ambient composition with known source profiles. The method asks: if we already know the chemical fingerprints of the sources in this area, what combination of these profiles best explains the ambient sample?

CMB is powerful when source profiles are representative and the source list is complete. It can be applied even with a relatively small number of ambient samples because the source profiles are provided as input. This makes it especially useful in well-characterized areas where local traffic, dust, industrial, biomass burning, or other source profiles are available.

The main limitation is that the method depends heavily on the quality of the source profiles and the source list. If an important source is missing, if the profile has changed during atmospheric transport, or if two sources have very similar profiles, the results can become uncertain or misleading.

9. PMF: what it says and how it works

Positive Matrix Factorization is one of the most widely used receptor models for PM source apportionment. PMF decomposes the measured data matrix into a contribution matrix and a profile matrix, plus residuals. In simple terms, it tries to find source-like factors that reconstruct the measured concentrations as well as possible.

PMF has two important features. First, it includes measurement uncertainties, so data points with higher uncertainty have less influence on the solution. Second, it imposes non-negativity constraints, meaning that source profiles and source contributions cannot become negative. This is important because negative mass contributions are not physically meaningful.

PMF does not require complete source profiles beforehand, but the analyst must still choose the number of factors and interpret them. This interpretation can be challenging because sources may be mixed, factors may rotate, and several mathematically similar solutions may be possible. Diagnostics and external evidence are therefore essential.

10. PCA, CMB, and PMF: key differences

The three methods answer related but different questions. PCA asks which variables vary together and is best seen as an exploratory tool. CMB asks how much each known source contributes, using predefined source profiles. PMF asks which source-like factors can be extracted from the ambient data itself, while accounting for uncertainties and non-negativity.

The practical implication is that each method requires a different type of confidence. For PCA, the analyst must be careful not to over-interpret statistical components. For CMB, the analyst must trust that the source profiles are correct and representative. For PMF, the analyst must evaluate whether the factor number, profiles, residuals, and temporal behavior make atmospheric sense.

Often, the methods are complementary. PCA can help explore the dataset, CMB can test known-source hypotheses, and PMF can provide a flexible source apportionment framework when complete profiles are not available.

Key comparison: PCA finds patterns, CMB uses known profiles, and PMF estimates unknown profiles and

contributions from the measured data.

11. A practical workflow for conventional SA

A robust source apportionment study begins with study design. Site selection, sampling frequency, chemical species, and time coverage should be chosen based on the research question. A study focused on traffic may need roadside and background sites, while a study focused on regional transport may need longer time coverage and meteorological analysis.

After measurement, the data must be prepared carefully. This includes quality assurance, treatment of missing and below-detection-limit values, calculation of uncertainties, and screening for outliers. These steps are not simply technical details: they strongly influence the model solution.

After the model is run, the results must be interpreted and evaluated. Source profiles should be checked against known tracers, source contributions should be compared with time patterns and meteorology, and diagnostics should be used to identify unstable or unrealistic solutions.

12. What data quality and diagnostics matter most?

Data quality determines the reliability of the final source apportionment. Species should be included only when they provide meaningful information and have realistic uncertainties. Variables that are mostly below detection limits, unstable, or poorly measured can distort the solution.

Before modeling, it is useful to examine time series, scatter plots, correlations, ion balance, and mass closure. During modeling, diagnostics such as residuals, Q values, factor stability, bootstrap tests, and displacement tests help evaluate whether the solution is robust. After modeling, source tracers, expected ratios, seasonality, wind direction, and trajectory analysis can support or challenge the proposed interpretation.

For PMF in particular, factor assignment is often the most subjective step. A mathematically valid factor is not automatically a real-world source. The analyst must connect the factor to atmospheric evidence.

Practical warning: A clean-looking model output is not enough. Always examine the data treatment, uncertainty estimates, residuals, and atmospheric plausibility of the factor interpretation.

13. Why move toward real-time source apportionment?

Offline source apportionment can provide strong scientific evidence, but it is usually too slow for rapid response. Pollution episodes can occur and disappear before laboratory analysis and modeling are complete. Real-time source apportionment addresses this limitation by providing source-resolved information shortly after measurement.

High-time-resolution measurements allow the analyst to see short events and diurnal patterns. For example, morning traffic peaks, evening residential heating, dust events, fireworks, industrial plumes, or rapid changes in secondary aerosol formation may be identified more clearly than in daily filter samples.

The main advantage is not only speed but also temporal detail. Real-time SA helps connect sources to specific events, meteorology, and human activities. However, it requires stable instruments, automated quality control, and well-defined model settings.

14. The instruments behind real-time SA

Real-time source apportionment requires online instruments that measure complementary parts of the PM composition. A single instrument does not characterize all PM components, so multi-instrument setups are used. The AXA setup combines an ACSM, an Xact or XRF-based elemental monitor, and an Aethalometer.

The ACSM measures non-refractory aerosol components such as organics, sulfate, nitrate, ammonium, and chloride. The Xact provides high-time-resolution elemental data, which are useful for identifying sources such as dust, industry, traffic wear, and combustion. The Aethalometer measures black carbon and can help separate liquid-fuel and solid-fuel combustion contributions.

Together, these instruments provide a much broader view of PM composition than any one instrument alone. Their data streams can be processed and used in real-time or near-real-time source apportionment workflows.

15. How can real-time source apportionment work?

A practical real-time SA workflow usually begins with automated measurement and processing. Online instruments produce concentration data, which are checked, synchronized, and formatted for modeling. The first output is chemical composition, not source contribution.

The next step is automated source apportionment. Depending on the framework, each instrument may be modeled separately, or multiple instruments may be handled within a combined system. For example, organics, elements, and black carbon can be apportioned in instrument-specific steps and then recombined to give a more complete PM source picture.

The final output is a continuously updated source contribution dataset. In most applications, the real-time model is configured using optimized conventional or supervised source apportionment results. In other words, conventional SA is often used to train, initialize, or constrain the real-time workflow.

Real-time workflow: Measure and process data, perform instrument-specific or combined automated SA, then recombine outputs into continuously updated source contributions.

16. Real-time model logic and implementation choices

Real-time SA is automated, but it is not automatic in the sense of being free from assumptions. The user must define the number and type of sources, the constraints, the treatment of uncertainties, and the rules for data acceptance. Once these settings are established, the model can operate with minimal intervention.

Constrained or semi-constrained approaches are often used so that source profiles remain physically meaningful. Moving-window methods can allow the model to adapt to recent data, while other approaches may expand the dataset as new measurements arrive. Each strategy involves a trade-off between stability and adaptability.

Because the system runs continuously, operational features are important: automatic quality checks, backups, flags for missing or suspicious data, and clear reporting outputs. A real-time SA system should be scientifically reliable and operationally robust.

17. What does real-time SA add scientifically?

Real-time SA adds scientific value by making source dynamics visible. Sources do not remain constant throughout the day or season. Traffic, cooking, residential heating, dust resuspension, industrial activity, and secondary aerosol formation can all vary on short timescales.

High-time-resolution source contributions allow researchers to isolate pollution episodes and study how source contributions change during them. This is particularly useful for evaluating the causes of exceedances, identifying transient sources, and understanding the response of PM composition to meteorology or control measures.

Recent applications using the AXA setup show that real-time systems can produce results comparable to optimized source apportionment when properly configured. This supports their use in urban monitoring, environmental agency workflows, and rapid air-quality assessment.

18. Conventional versus real-time SA: when should each be used?

Conventional and real-time SA are best viewed as complementary. Conventional SA is the foundation: it provides carefully validated source profiles, factor interpretations, and uncertainty assessments. Real-time SA builds on this foundation to provide continuous, rapid source-resolved information.

For retrospective research, long-term trend analysis, or detailed method development, conventional approaches are often preferable. For operational monitoring, event detection, and rapid interpretation, real-time approaches are more useful. A strong monitoring strategy can use both: conventional SA to optimize and validate, and real-time SA to deploy the optimized framework continuously.

The central question is not which approach is better, but which approach is appropriate for the decision that must be made.

19. Take-home messages

Source apportionment converts PM composition measurements into source-specific information. It relies on the fact that sources have different chemical fingerprints and that measured ambient PM can be treated as a mixture of source contributions.

PCA, CMB, and PMF each have a different role. PCA is useful for exploring patterns, CMB is useful when reliable source profiles are available, and PMF is a flexible uncertainty-weighted method that estimates profiles and contributions from ambient data. All models require careful interpretation and validation.

Real-time SA extends these ideas into operational monitoring by using online instruments, automated data processing, and near-real-time modeling. The best practice is to understand the assumptions, optimize the method carefully, and use conventional and real-time approaches together.

Quick reference: model comparison

Feature	PCA	CMB	PMF
Main role	Exploratory pattern recognition	Quantification using known source profiles	Factorization of ambient data into profiles and contributions
Prior source profiles	Not required	Required	Not required, but constraints can be used
Uncertainty weighting	Usually not central	Yes	Yes, essential
Physical constraints	Limited; negative values possible	Depends on implementation	Non-negativity constraints
Typical use	Screening and data exploration	Well-characterized areas	Most common flexible receptor-model approach
Main risk	Over-interpreting components as sources	Wrong or incomplete source profiles	Rotational ambiguity, mixed factors, subjective

			interpretation

Glossary of key terms

Source apportionment: Identification and quantification of pollution sources contributing to measured concentrations.

Receptor model: A model that uses measurements at a receptor site to infer source contributions.

Source profile: The chemical fingerprint of a source, expressed through the relative abundance of measured species.

Source contribution: The amount of PM mass attributed to a source at a given time or over a defined period.

PCA: Principal Component Analysis; an exploratory method that identifies patterns of covariance.

CMB: Chemical Mass Balance; a receptor model that uses known source profiles to estimate contributions.

PMF: Positive Matrix Factorization; a receptor model that estimates profiles and contributions while accounting for uncertainties.

ME-2: Multilinear Engine; a flexible factorization framework that can include constraints or prior information.

AXA setup: A combined online measurement setup using ACSM, Xact/XRF, and Aethalometer data.

Real-time SA: Automated or semi-automated SA that produces updated source contributions soon after measurements are collected.

Review questions for students

- Why is chemical composition essential for PM source apportionment?
- What is the difference between a source profile and a source contribution?
- Why can PCA suggest source patterns but not always provide physically constrained source contributions?
- What information must be available before CMB can be applied reliably?
- Why are uncertainties central to PMF?
- What is rotational ambiguity in PMF, and why does it matter?
- Why are conventional and real-time SA better considered complementary rather than competing approaches?
- What are the roles of ACSM, Xact/XRF, and Aethalometer in an AXA real-time SA setup?